OGE-JEPA-TTC v6: Auditable Dense Event Prediction, Object Geometry, and a Local Garl-TTC Baseline

KriptaStudios Research Handoff

Reproducible local report from the `e-jepa-ttc` repository

Protocol revision: 30 July 2026

Abstract

This report describes version 6 of an event-camera time-to-collision research pipeline. The revision does not claim state of the art. It first reconstructs the historical Event-JEPA baseline exactly, separates that historical anchor from a matched architecture control, and implements staged tests for dense patch interaction, task-specific Attention Residuals, temporal delta memory, object geometry, ego-motion, and a local Garl-TTC replica. The historical checkpoint is reproduced byte-for-byte on its original cache and obtains 0.322892 s validation MAE, 0.584432 s RMSE, and 8.1554% mean relative error. FlowMimic-style global auxiliary losses are retained only as a negative result because they degrade multi-seed MAE.

A matched confirmation gives every Core arm the same checkpoint, samples, optimizer, maximum 40 epochs, and stopping rule. Dense Patch improves sequence-macro relative error from 16.129% to 15.210% and macro MAE from 0.701 to 0.628 s. Attention Residuals and Object-KDA do not improve the dense arm. Five-fold, three-seed grouped cross-validation nevertheless selects A0: A1 degrades the composite score by 1.47% and relative error by 0.99%, while improving MAE by only 0.41% at 2.16× latency. A subsequent bbox-ROI arm also fails its sequential gate. The official Benchmark-10 remains sealed; these local results are not a state-of-the-art claim.

1 Scope and claim boundary

The project estimates continuous TTC from asynchronous event streams and, conditionally, RGB and causal navigation. The long-term hypothesis is that preserving dense spatial evidence, isolating the target object, and embedding physical expansion in the output path can outperform regression from a pooled scene embedding. This hypothesis is not yet accepted by the evidence.

The data inventory is closed:

- EvTTC-32 is the only official TTC supervision source;
- eAP train-40 supplies a signed 12-sequence, 9/3 train/validation pilot for non-TTC event-only pretraining;
- CARLA DVS Looming [5] supplies synthetic positive TTC and censored negatives for pretraining and cross-domain transfer;
- Benchmark-10 is an external sealed evaluation set;
- pseudo-TTC is never treated as ground truth.

Table 1: Exact historical BASE validation reproduction, seed 7.

Quantity	Value
Train windows	7,835
Validation windows	2,040
Downstream checkpoint	epoch 26/30
SSL checkpoint	epoch 6/30
MAE	0.322892 s
RMSE	0.584432 s
Mean relative error	8.1554%

2 Historical evidence

2.1 Exact BASE reconstruction

The audited baseline consumes 10 event channels and 11 causal auxiliary channels. Its EventTubeletTransformer uses dimension 192, depth 6, six heads, and patch size 16. Final dense tokens are mean-pooled and passed through LayerNorm, a 192-to-96 MLP, GELU, dropout 0.1, and a scalar log-TTC output.

Prediction arrays are byte-equivalent to the original stored predictions, and all audited metric differences are zero at tolerance 10^{-12}. This model is named `B0_HISTORICAL_BASE_EXACT`. It is not reused as if it had been trained under the new object-cache protocol.

2.2 Negative FlowMimic result

These losses are removed from the active architecture. Their summary remains versioned to preserve negative evidence.

Table 2: Historical multi-seed MAE. Lower is better.

Arm	MAE (s)	Change
BASE	0.332715	–
Global alignment	0.369604	+11.1%
Synthetic inverse-TTC	0.432909	+30.1%
Both losses	0.435869	+31.0%

Table 3: Matched Core confirmation on 1,208/314 train/validation windows. All arms share a 40-epoch maximum and one early-stopping rule.

Arm	Best	Rel.	MAE	ms
A0 global	17	16.129%	0.701	8.98
A1 Dense	**20**	**15.210%**	**0.628**	17.15
A2 AttnRes	10	16.136%	0.653	16.70
K1 Object-KDA	7	16.960%	0.731	16.99

3 Matched architecture protocol

The learned Core arms are:

- `A0_MATCHED_GLOBAL`, the object-cache global control;
- `A1_MATCHED_DENSE_BLOCK`, delayed pooling;
- `A2_MATCHED_DENSE_ATTNRES`, depth routing by task;
- `K1_OBJECT_KDA`, temporal delta memory.

The fairness audit requires identical sample IDs, backbone hash, common head hash, train and validation counts, optimizer, learning rate, weight decay, maximum epochs, log-TTC loss, and early stopping. Actual completed epochs may differ only through the shared stopping rule.

Core and Garl write separate stage roots:

```
artifacts/runs/evttc32_architecture_v4_<split>_<mode>/
  core/fold-<n>/matrix_summary.json
  garl/fold-<n>/matrix_summary.json
```

This prevents the previous summary collision while retaining individual run directories.

The eight-epoch screen ranked A1 incorrectly because its best checkpoint appears at epoch 20. A1 completed 26 epochs while A0 completed 23 under the same rule; A0 did not receive more optimization. A1 improves relative error by 5.70% and MAE by 10.45%, with a 1.91-fold latency cost. A2 and K1 are not promoted in their current forms.

The predeclared grouped-CV gate completed all 30 A0/A1 runs. A0 obtains 0.58452 ± 0.00853 score and $30.25\% \pm 0.52$ relative error, compared with 0.59312 ± 0.00349 and $30.55\% \pm 0.06$ for A1. The later `R1_MATCHED_BBOX_ROI` test uses ground-truth boxes only to pool dense tokens. Across five seed-7 folds it degrades A0 by 2.90% in score, 2.74% in relative error, and 4.55% in MAE, so seeds 13 and 21 are not spent. Region selection alone does not measure apparent expansion.

4 Dense spatial and temporal design

4.1 Patch Policy

Patches from one timestamp communicate bidirectionally before any temporal compression. Causality exists between time blocks, not between arbitrary patches inside a frame. Pooling before target localization is forbidden in the dense arm. This implements the dense representation principle motivated by Patch Policy [1].

4.2 Attention Residuals

Mask, motion, geometry, and risk tasks independently retrieve features from embedding and block outputs. Keys are RMS-normalized so a layer cannot dominate by scale; the mixed values remain the original features. A permanent greater-than-90% weight on one layer is treated as collapse.

4.3 Temporal delta memory

The KDA candidate follows the delta-recurrence principle described by Kimi K3 [6], but is restricted to the temporal axis. Spatial interaction is resolved first. KDA is optional and must show either a precision gain, a material memory reduction, or a longer effective horizon.

5 Local Garl-TTC baseline

Garl-TTC is implemented from the public source associated with the eAP work [3]. The audited local contract uses three timestamps at 100 ms cadence, RGB endpoints, two event intervals, 20 time planes per interval, and a common 128-by-128 endpoint square crop. RGB and event ResNet-50 branches can be trained for direct regression or learned height ratio (LHR). Late fusion initializes its branches from the corresponding unimodal LHR checkpoints. A foreground decoder produces four 256-by-256 logits during training and is absent from inference.

EvTTC does not expose the original eAP 3D visible-height target. The local adapter therefore supervises visible box height after the shared crop. Results must be reported as a local architectural replica, not official eAP parity. In the bounded local screen, RGBE-LHR early fusion is best at 36.52% sequence-macro relative error. This is a discard screen, not parity evidence: the public configuration trains for 50 epochs and constructs late fusion from pretrained unimodal LHR branches.

6 Object geometry

For apparent scale s, inverse-TTC at the current endpoint is

$$q_t = \frac{s_t/s_{t-\Delta t} - 1}{\Delta t}. \quad (1)$$

Height, square-root area, and affine expansion all use the latest valid pair. The affine solver separates translation, radial expansion, and in-plane rotation through weighted least squares. Event contrast evaluates a bounded constant-approach warp only inside the object support. Auxiliary metadata channels are excluded from the event calculation.

The training-free bbox oracle reports each expert and a deterministic confidence mixture. It is an upper-bound diagnostic and not a bbox-free model.

A separate causal bbox-scale estimator fits a local derivative over at most 21 past detections and a train-only log-affine calibration. It covers 311 of 314 validation windows. Using A1 only for the three invalid rows changes macro relative error from 15.210% to 14.790%, but worsens the composite selection score from 0.30543 to 0.31144 because of RMSE. It therefore fails the declared 5% gate and also uses more history than the three-frame neural control.

We also ported the public STRTTC chain: nearest linear time surface, contour gradients, local event-plane normal flow, RANSAC, and the three-parameter solver [4]. The bounded causal screen solves only 27/40 samples; its successful rows have 112.96% sequence-macro relative error. Failures are reported as missing coverage, not silently removed from a full-set claim.

7 Ego-motion

The EvTTC format states that integrated navigation attitude, position, and velocity live in a world coordinate system [2]. Local auditing shows velocity components ordered north, east, up and heading in degrees. The heading agrees with geodetic displacement. Velocity is transformed through the documented rigid chain

$$T_{\text{event}\leftarrow\text{nav}} = T_{\text{event}\leftarrow\text{RGB}} T_{\text{RGB}\leftarrow\text{LiDAR}} T_{\text{LiDAR}\leftarrow\text{nav}}. \quad (2)$$

The lever-arm velocity of the event-camera origin is included when heading changes.

For a past point with causal depth, full planar camera-pose compensation uses

$$P_c = R_{c\leftarrow p} P_p - \Delta C_c. \quad (3)$$

The boxes are reprojected with event-camera intrinsics. Ego closing contributes $q_{\text{ego}} = v_z/Z$. Translation is not observable from velocity alone: depth is required. Consequently, official EvTTC distance is allowed only for an oracle or teacher. A deployable arm must replace it with causal predicted depth and be re-evaluated.

8 Experimental stages

The checkpoint score combines sequence-macro relative error, normalized RMSE, high-TTC error, and low-TTC safety error. Ties are resolved by worst sequence, RMSE, seed variance, and latency. Smoke and Screen are closed; historical Confirm promotes A1 only, while completed grouped CV selects A0. Grouped CV compares A0/A1 from a common random EventTubelet initialization because reusing the historical SSL checkpoint would expose some validation sequences during pretraining. The checkpoint-initialized historical Confirm remains a separate experiment.

Table 4: Evidence hierarchy.

Stage	Purpose
Smoke	Integration only, never promotion
Screen	304/80 windows, seed 7, up to 8 epochs
Confirm	1,208/314 windows, up to 40 epochs
Grouped CV	Five folds, seeds 7, 13, 21 for A0/A1
Sequential ablation	Five folds, seed 7 before further seeds
External	Frozen Benchmark-10 inference

TargetQuery, predicted masks, high-resolution refinement, learned routing, bounded residual, and uncertainty remain behind a hard gate: bbox-GT geometry must first outperform BASE under the declared protocol.

9 Efficiency and storage

The consumer hardware target has approximately 12.8 GB VRAM and 32 GB RAM. Training uses BF16, pinned memory, persistent workers, prefetch, and gradient accumulation. The Garl ResNet-50 screen uses batch 24; the foreground arm uses microbatch 4 with accumulation 6 for the same effective batch. The geometric solver remains FP32.

No full eAP or CARLA voxel cache, full-resolution SAM logits, all-layer DINO cache, or bulk TAR extraction is permitted. CARLA's 7.692 billion events are read by memory-mapped causal windows without a second raw copy. A run retains at most `best`, `last`, and `weights_only`. Core and Garl caches are compressed and stage-specific.

The CARLA full profile creates 12,020/4,457/4,297 lazy train/validation/test pairs, uses BF16, batch 24 with accumulation 2, eight persistent workers, fused AdamW, warm-up/cosine decay, EMA, clipping, and 8/6 early stopping with a 30-epoch maximum. The measured best probe processed 8.46 pair observations per second; batch 16/32/48/96 and 6/12-worker profiles did not improve it because CPU/SSD voxelization, rather than VRAM, was limiting. This projects to 32.5 minutes per epoch and 16.2 hours at the hard maximum.

The eAP pilot reads HDF5 event windows on demand

Table 5: eAP-12 transfer screen (positive means lower error/score).

Init	Model	ΔRTE	ΔMAE	Δscore
SSL	A0	-2.58%	-1.13%	-3.43%
SSL	A1	+0.54%	+6.33%	-0.02%
Geo	A0	+3.66%	+4.30%	+2.36%
Geo	A1	+6.57%	+7.95%	+6.47%

through `ms_to_idx`; it neither opens the 118 GiB RGB payload nor materializes a global voxel cache. The signed selection contains nine training and three validation sequences. `eAP-SSL` predicts dense future embeddings, whereas `eAP-Geo` adds projected 3-D-box position and size, radial closing, apparent expansion, and patch objectness. Neither arm uses reconstructed TTC as a target. Both use the same EventTubelet encoder, seed, windows, batch, and optimization budget. The full profile uses BF16, batch 24 with accumulation 2, eight persistent workers, fused AdamW, TF32, clipping, EMA, cosine decay, and 8/6 early stopping up to 30 epochs. It switches to a signed, label-independent 32/8 split over all 40 sequences (16,384/4,096 windows). The three pilot validation sequences remain held out; five more are selected by a salted hash of sequence ID. Utility is measured only by matched transfer to A0 and A1 on EvTTC.

The three-epoch pilot selected epoch three for both arms. SSL obtained latent validation loss 0.002358 in 11.84 minutes; Geo obtained total loss 0.087108, JEPA loss 0.007624, and patch IoU 0.2867 in 13.06 minutes. Table 5 reports relative improvements over each architecture's paired random control on folds 0/1, seed 7.

10 Limitations and next steps

Dense Patch improves one matched split but loses completed grouped-CV multi-seed evaluation. Bbox-ROI pooling also loses its five-fold seed-7 gate. The local Garl screen is shorter than the official 50-epoch, branch-pretrained recipe. STRTTC has incomplete coverage and translation compensation is not deployable until depth is predicted. eAP train-40 is complete but has no official TTC labels. CARLA is synthetic, quantized at 10 ms, lacks temporal boxes, and provides positive TTC only to approximately 3.85 s. Benchmark-10 has not been opened.

The two-epoch CARLA SSL smoke reduced validation loss from 0.02563 to 0.02247 without collapsed dimensions. A 16-pair synthetic test contract produced 0.02195. However, matched fold-0 transfer worsens A0 relative error by 1.72% for SSL and by 17.3% for the synthetic-TTC auxiliary arm. CARLA is therefore retained as negative evidence rather than promoted. Geo improves A0 RTE and MAE on both screened folds, opening the predeclared 12-to-40 scaling gate. A1 Geo improves RTE on both folds but MAE on only one. The RTE cluster-bootstrap intervals still cross zero; full eAP-40 pretraining and five-fold, three-seed EvTTC confirmation remain outstanding. An explicit expansion/FoE path remains a later candidate. These restrictions preclude a current SOTA or safety claim.

References

[1] Anonymous. Patch policy: Efficient embodied control via dense visual representations. *arXiv preprint arXiv:2607.18236*, 2026.

[2] Jinghang Li et al. Evttc: An event camera dataset for time-to-collision estimation. *arXiv preprint arXiv:2412.05053*, 2024.

[3] Jinghang Li, Shichao Li, Qing Lian, Peiliang Li, Xiaozhi Chen, and Yi Zhou. Toward deep representation learning for event-enhanced visual autonomous perception: the eap dataset. *arXiv preprint arXiv:2603.16303*, 2026.

[4] Jinghang Li, Bangyan Liao, Xiuyuan Lu, Peidong Liu, Shaojie Shen, and Yi Zhou. Event-aided time-to-collision estimation for autonomous driving. In *European Conference on Computer Vision*, 2024.

[5] Fabian Schubert et al. Carla simulator collision scenario dvs sequences with and without traffic and crossings, 2025.

[6] Kimi Team. Kimi k3: Open frontier intelligence. *arXiv preprint arXiv:2607.24653*, 2026.